RS62 Yield, Efficiency & Cross-Section Comparison

Existing vs. Latest Results

October 28, 2026

Single Differential Cross-Section Formalism ($\mathrm{d}\sigma/\mathrm{d}p_T$)

LH_2 (pp) Single Differential Cross Section:

$$\frac{\mathrm{d}\sigma_{pp}}{\mathrm{d}p_T} = \frac{1}{\epsilon_{\mathrm{acc}}\mathcal{L}\Delta p_T}\left[\frac{Y^{\mathrm{LH2}}_{\mathrm{total}} - Y^{\mathrm{LH2}}_{\mathrm{mixed}}}{\langle\epsilon^{\mathrm{LH2}}_{\mathrm{signal}}\rangle} - \frac{I_{\mathrm{LH2}}}{I_{\mathrm{flask}}}\left(\frac{Y^{\mathrm{flask}}_{\mathrm{total}} - Y^{\mathrm{flask}}_{\mathrm{mixed}}}{\langle\epsilon^{\mathrm{flask}}_{\mathrm{signal}}\rangle}\right)\right] \tag{1}$$

LD_2 (pd) Single Differential Cross Section:

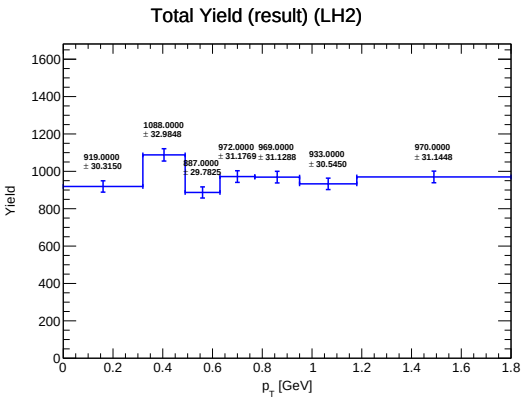
$$\begin{aligned}\frac{\mathrm{d}\sigma_{pd}}{\mathrm{d}p_T} = \frac{1}{\epsilon_{\mathrm{acc}}\mathcal{L}\Delta p_T}\Bigg(&\left[\frac{Y^{\mathrm{LD2}}_{\mathrm{total}} - Y^{\mathrm{LD2}}_{\mathrm{mixed}}}{\langle\epsilon^{\mathrm{LD2}}_{\mathrm{signal}}\rangle} - \frac{I_{\mathrm{LD2}}}{I_{\mathrm{flask}}}\left(\frac{Y^{\mathrm{flask}}_{\mathrm{total}} - Y^{\mathrm{flask}}_{\mathrm{mixed}}}{\langle\epsilon^{\mathrm{flask}}_{\mathrm{signal}}\rangle}\right)\right] \\ &- \frac{T_{HD}}{T_{HH}}\frac{I_{\mathrm{LD2}}}{I_{\mathrm{LH2}}}\left[\frac{Y^{\mathrm{LH2}}_{\mathrm{total}} - Y^{\mathrm{LH2}}_{\mathrm{mixed}}}{\langle\epsilon^{\mathrm{LH2}}_{\mathrm{signal}}\rangle} - \frac{I_{\mathrm{LH2}}}{I_{\mathrm{flask}}}\left(\frac{Y^{\mathrm{flask}}_{\mathrm{total}} - Y^{\mathrm{flask}}_{\mathrm{mixed}}}{\langle\epsilon^{\mathrm{flask}}_{\mathrm{signal}}\rangle}\right)\right]\Bigg)\end{aligned} \tag{2}$$

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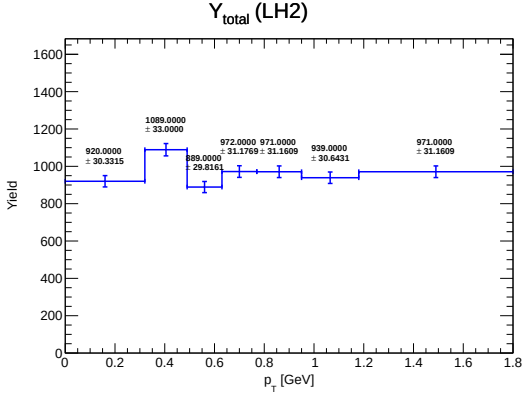
1. Yield Distributions
2. Efficiency Distributions
3. Corrected & Subtracted Yields
4. Cross Sections & Ratios
5. Systematic Uncertainties

LH_2 Target: Total Yield (Y_total_LH2.pdf)

Existing Results

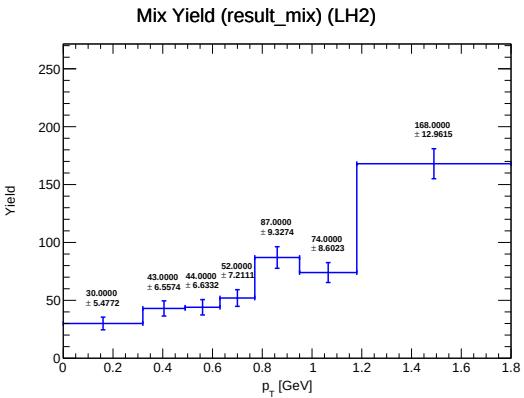


Latest Results

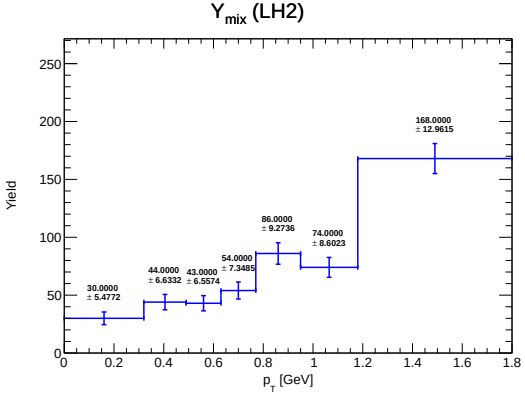


LH_2 Target: Mixed Yield (Y_mix_LH2.pdf)

Existing Results

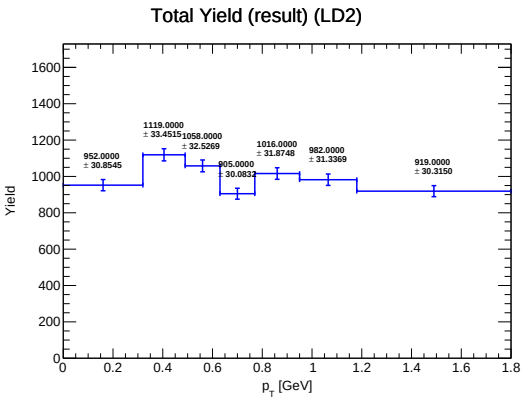


Latest Results

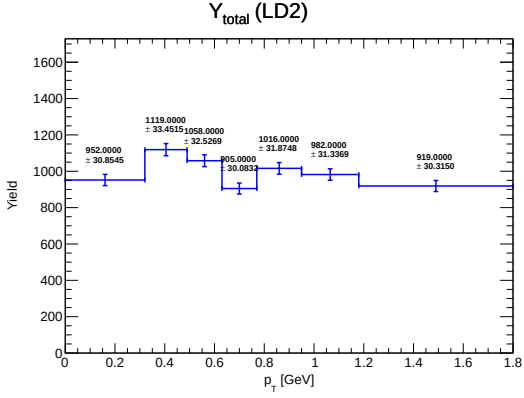


LD_2 Target: Total Yield (Y_total_LD2.pdf)

Existing Results

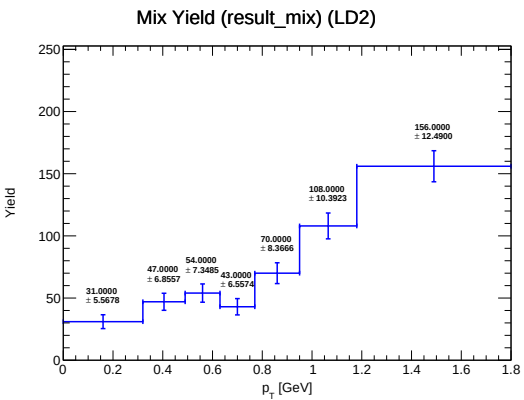


Latest Results

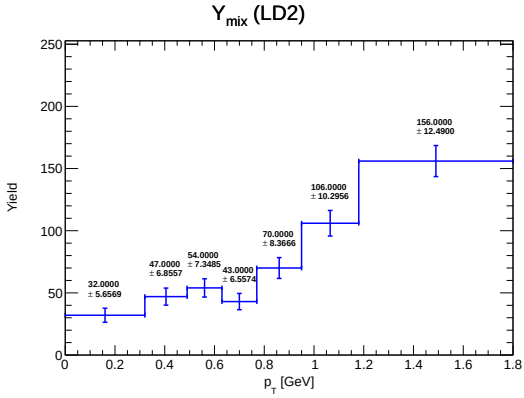


LD_2 Target: Mixed Yield (Y_mix_LD2.pdf)

Existing Results

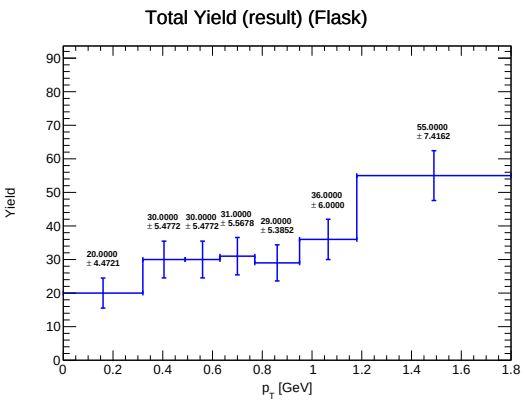


Latest Results

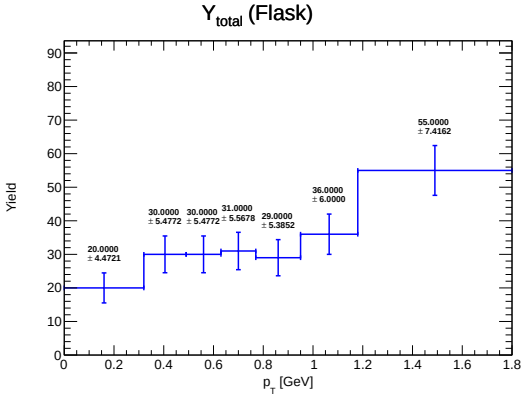


Empty Flask: Total Yield (Y_total_Flask.pdf)

Existing Results

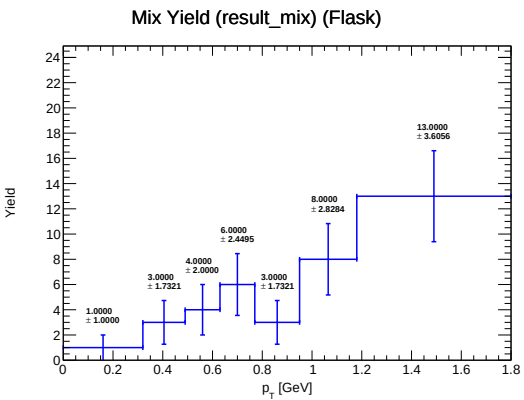


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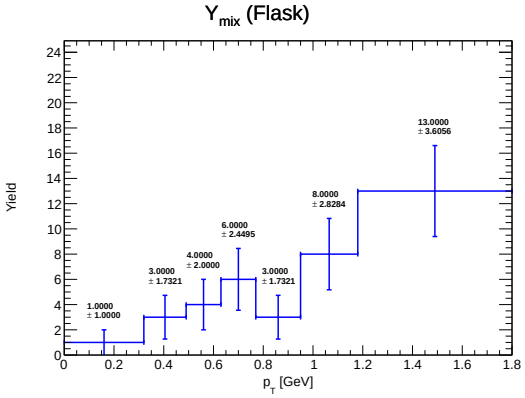


Empty Flask: Mixed Yield (Y_mix_Flask.pdf)

Existing Results

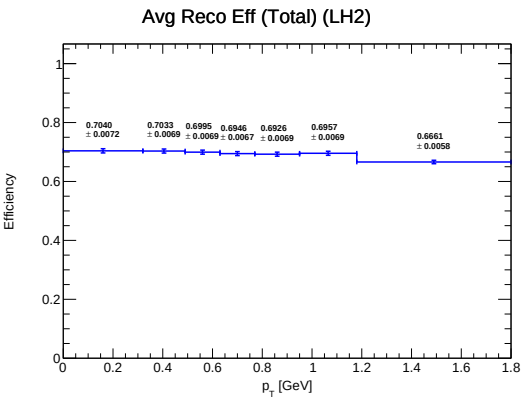


Latest Results

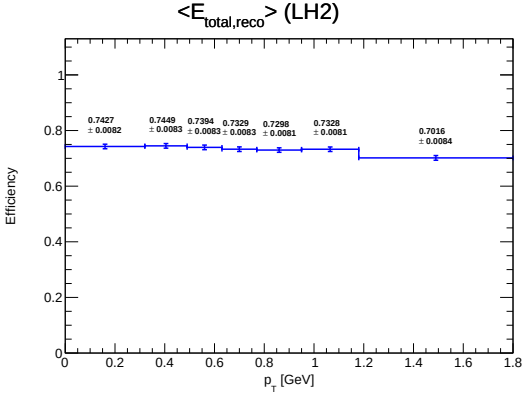


LH_2 Target: Total Reco Efficiency (E_total_reco_LH2.pdf)

Existing Results

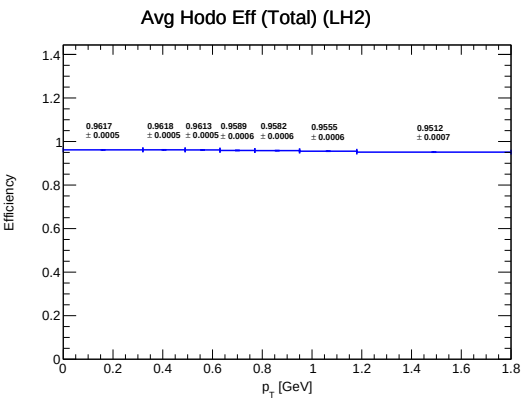


Latest Results

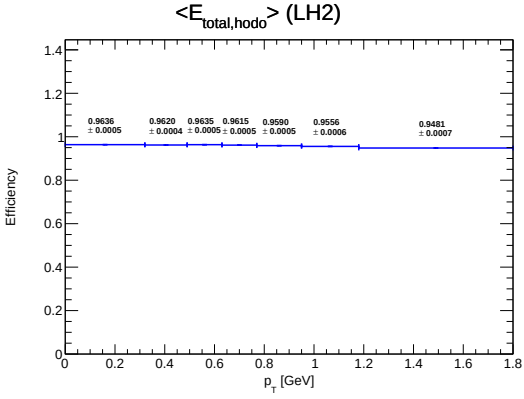


LH_2 Target: Total Hodo Efficiency (E_total_hodo_LH2.pdf)

Existing Results

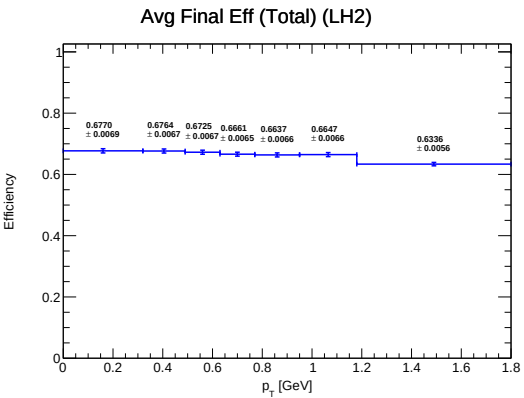


Latest Results

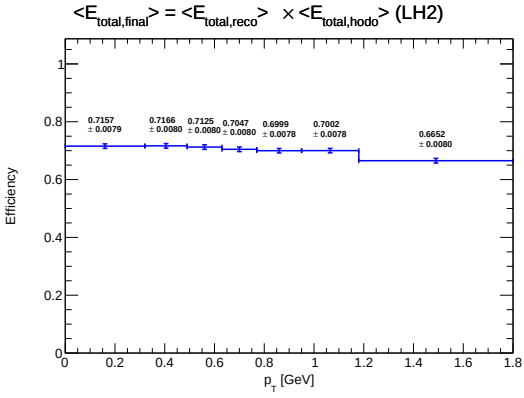


LH_2 Target: Total Final Efficiency (E_total_final_LH2.pdf)

Existing Results

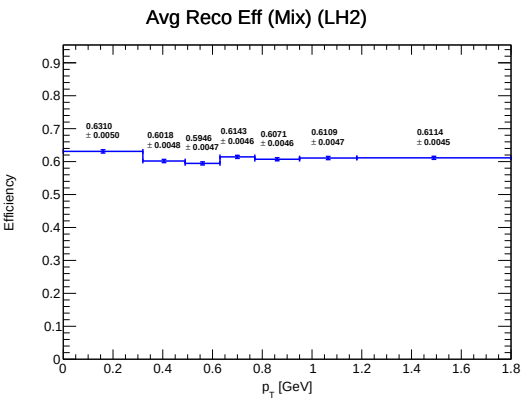


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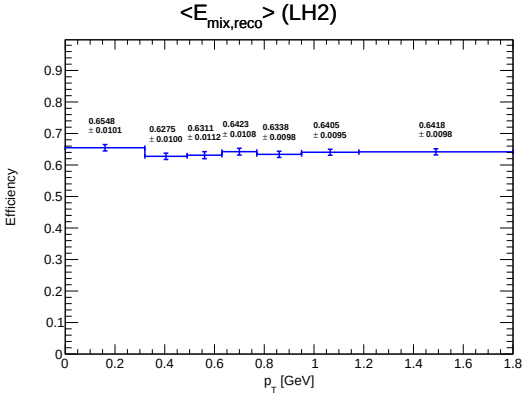


LH_2 Target: Mixed Reco Efficiency (E_mix_reco_LH2.pdf)

Existing Results

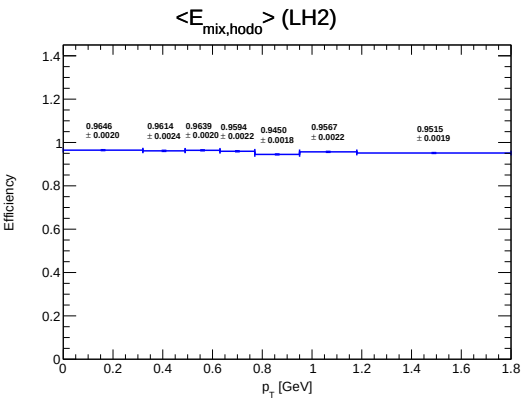


Latest Results

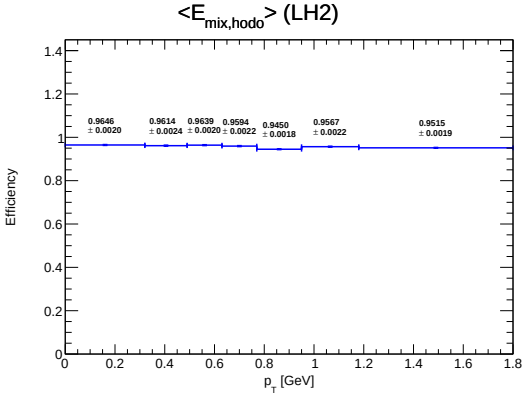


LH_2 Target: Mixed Hodo Efficiency (`E_mix_hodo_LH2.pdf`)

Existing Results

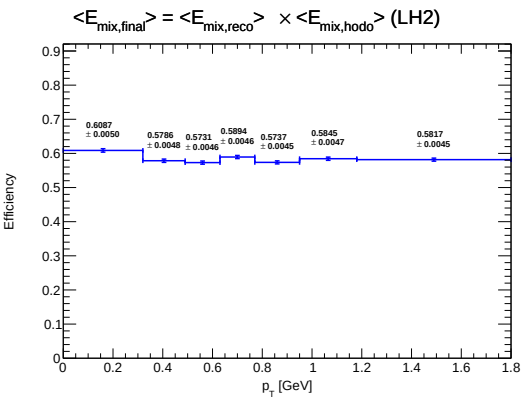


Latest Results

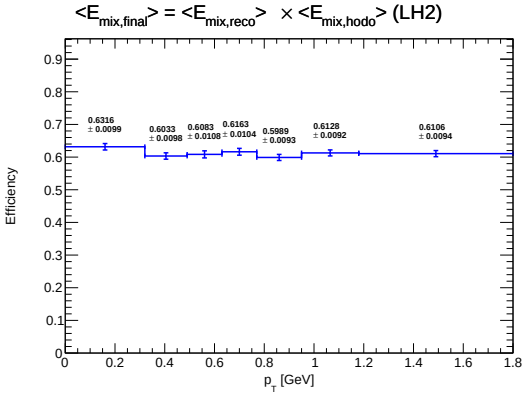


LH_2 Target: Mixed Final Efficiency (E_mix_final_LH2.pdf)

Existing Results

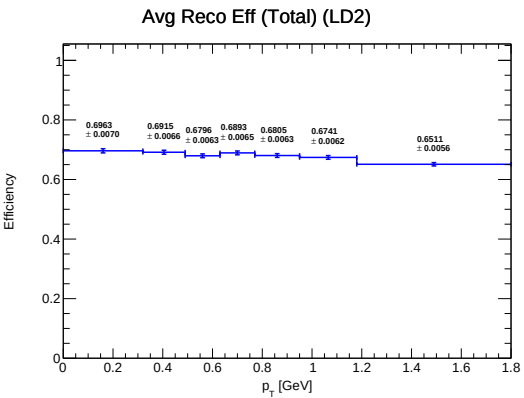


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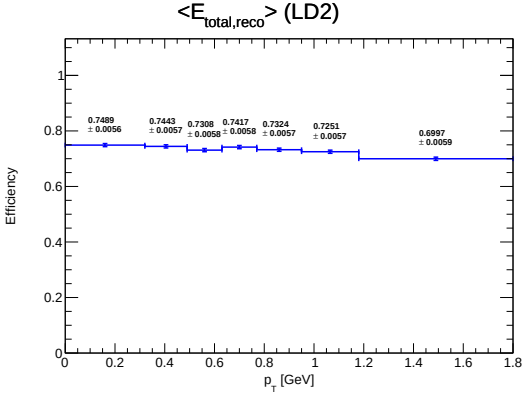


LD_2 Target: Total Reco Efficiency (E_total_reco_LD2.pdf)

Existing Results

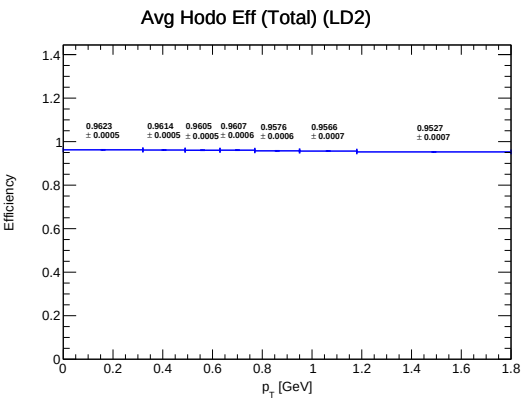


Latest Results

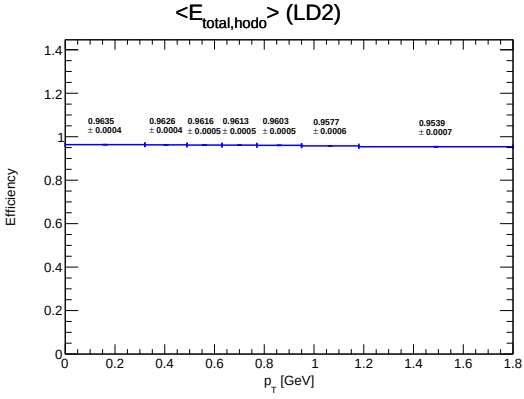


LD_2 Target: Total Hodo Efficiency (E_total_hodo_LD2.pdf)

Existing Results

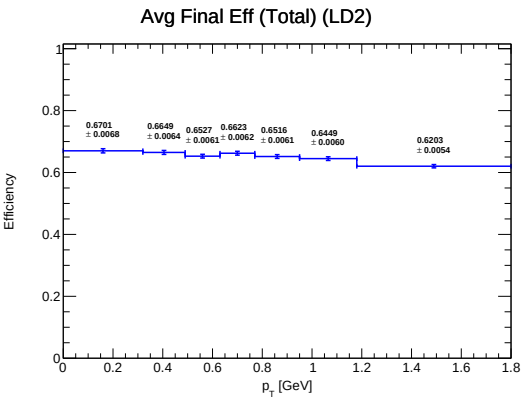


Latest Results

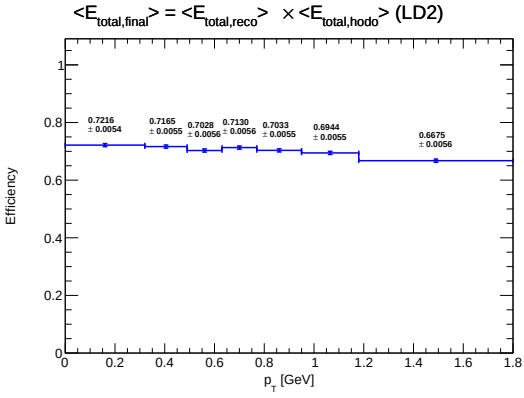


LD_2 Target: Total Final Efficiency (`E_total_final_LD2.pdf`)

Existing Results

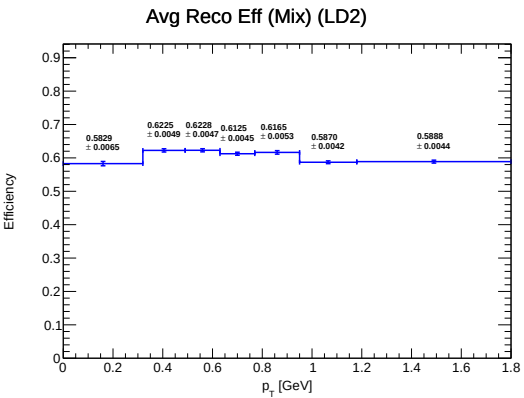


Latest Results

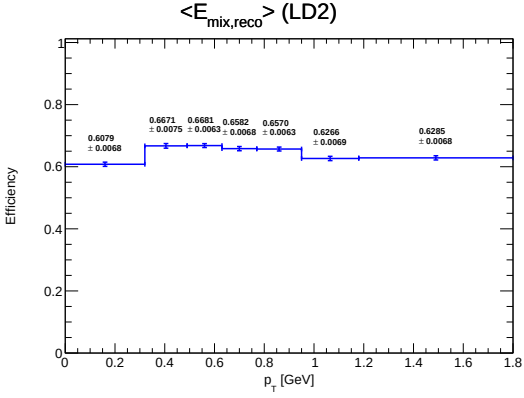


LD_2 Target: Mixed Reco Efficiency (E_mix_reco_LD2.pdf)

Existing Results

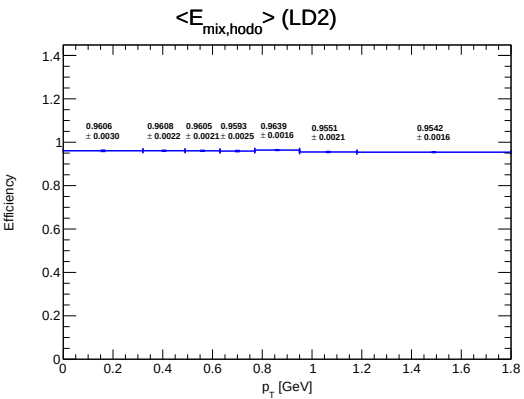


Latest Results

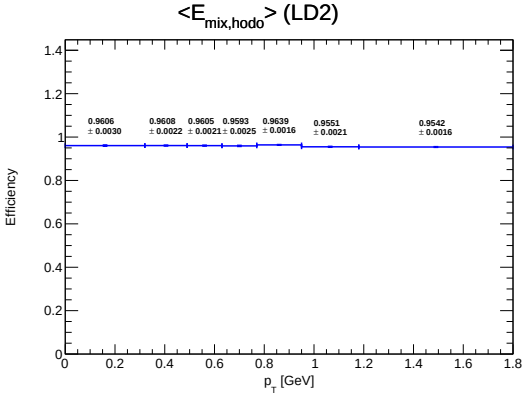


LD_2 Target: Mixed Hodo Efficiency (E_mix_hodo_LD2.pdf)

Existing Results

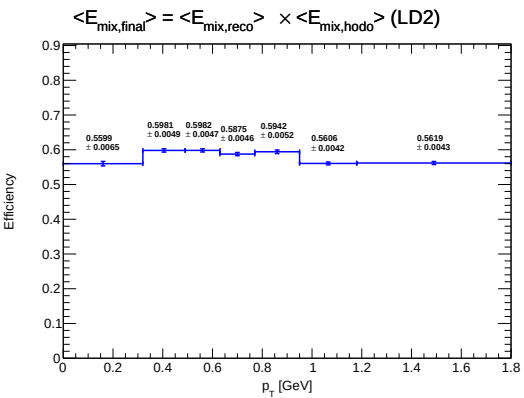


Latest Results

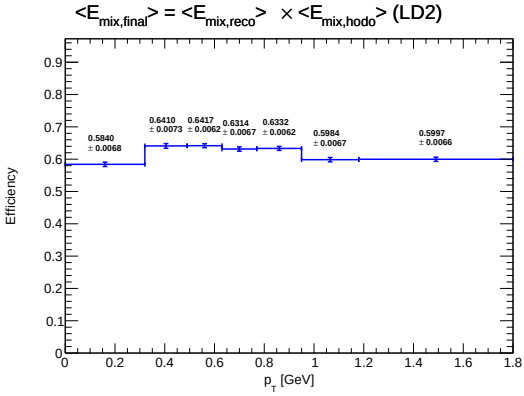


LD_2 Target: Mixed Final Efficiency (E_mix_final_LD2.pdf)

Existing Results

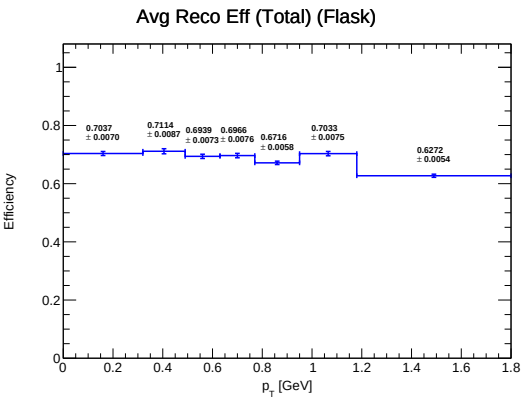


Latest Results

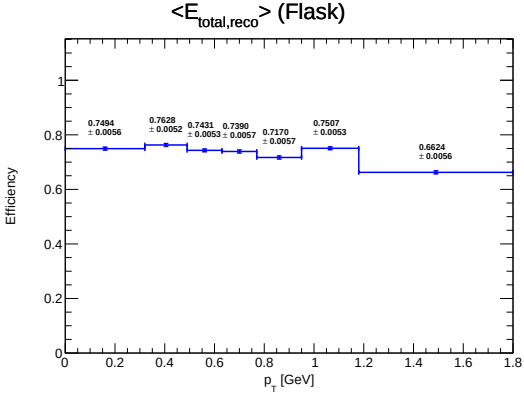


Empty Flask: Total Reco Efficiency (E_total_reco_Flask.pdf)

Existing Results

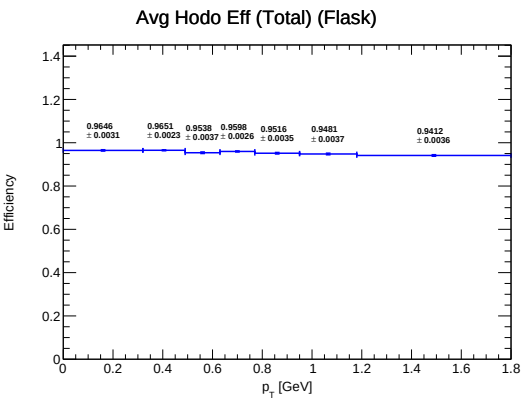


Latest Results

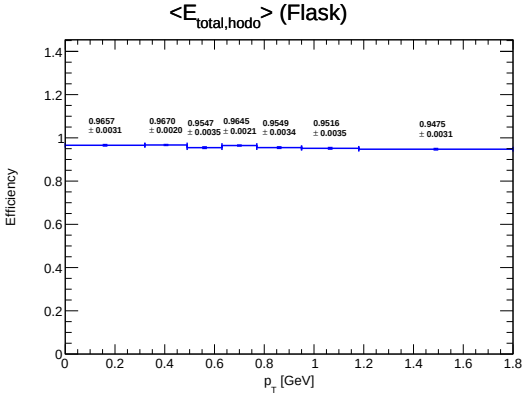


Empty Flask: Total Hodo Efficiency (E_total_hodo_Flask.pdf)

Existing Results

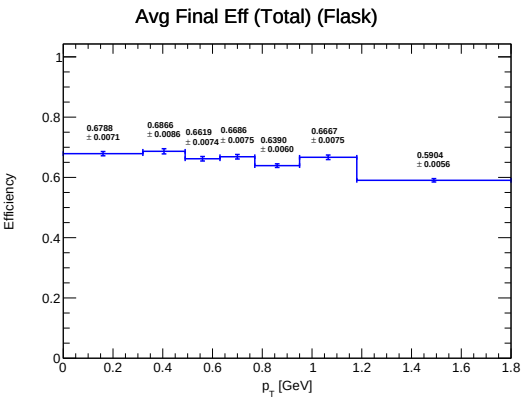


Latest Results

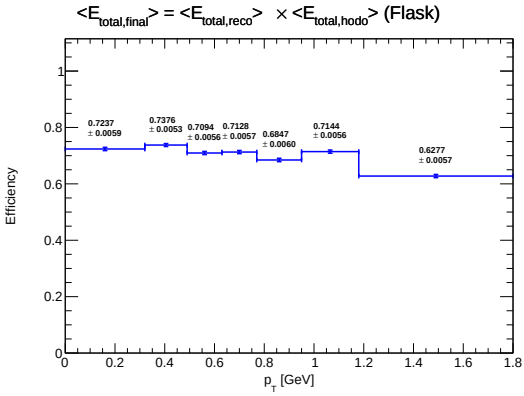


Empty Flask: Total Final Efficiency (E_total_final_Flask.pdf)

Existing Results

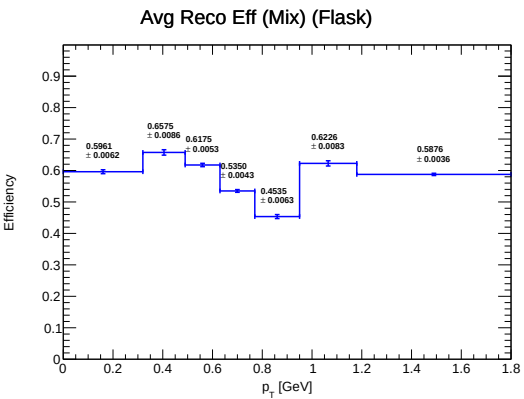


Latest Results

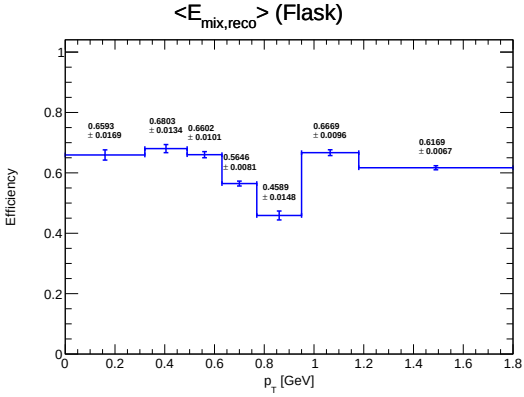


Empty Flask: Mixed Reco Efficiency (E_mix_reco_Flask.pdf)

Existing Results

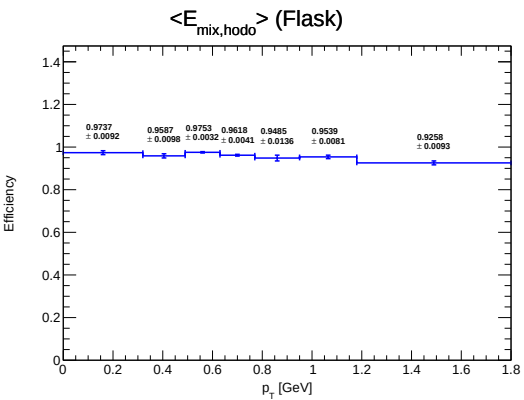


Latest Results

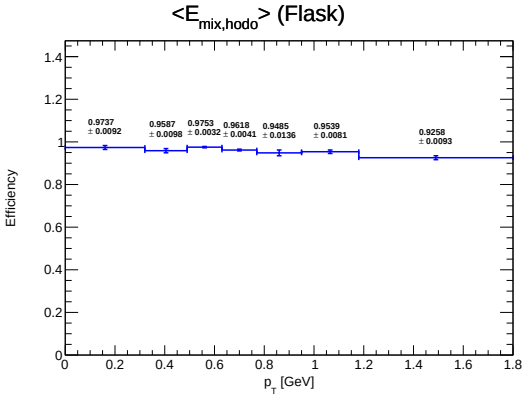


Empty Flask: Mixed Hodo Efficiency (E_mix_hodo_Flask.pdf)

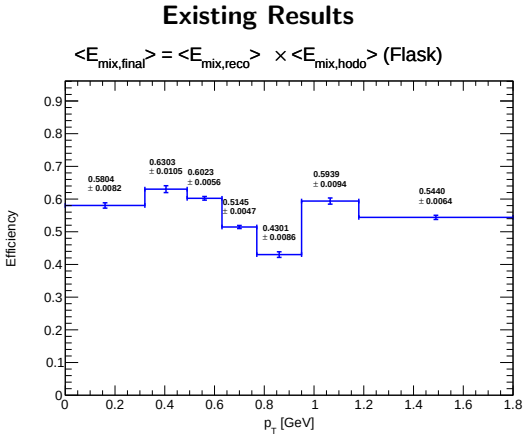
Existing Results

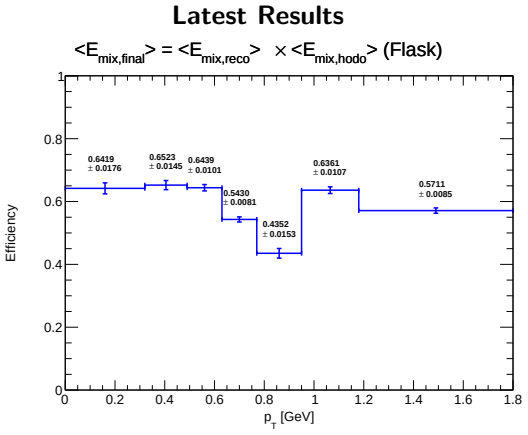


Latest Results



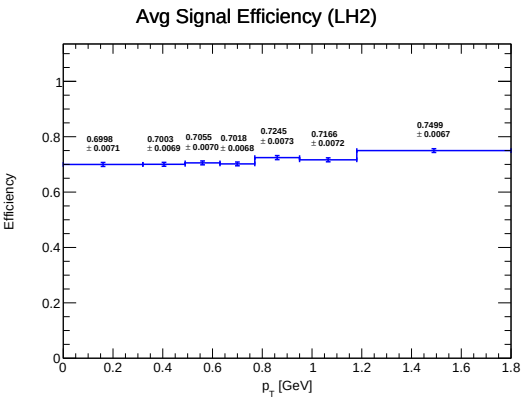
Empty Flask: Mixed Final Efficiency (E_mix_final_Flask.pdf)



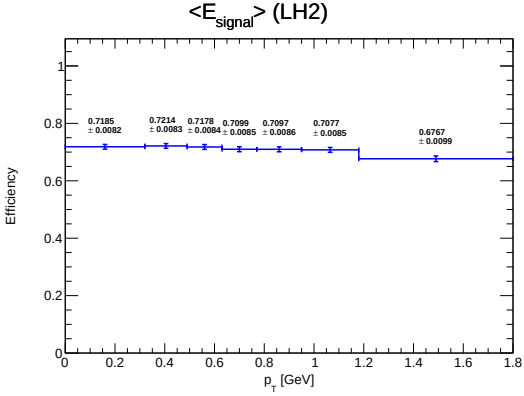


LH_2 Target: Final Signal Efficiency (E_final_signal_LH2.pdf)

Existing Results

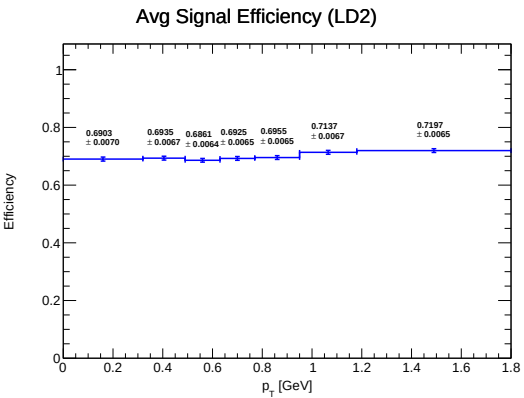


Latest Results

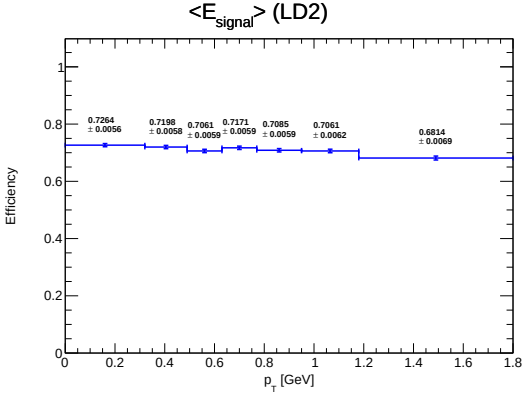


LD_2 Target: Final Signal Efficiency (`E_final_signal_LD2.pdf`)

Existing Results

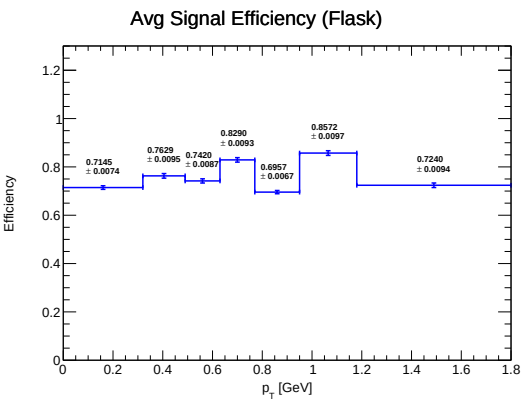


Latest Results

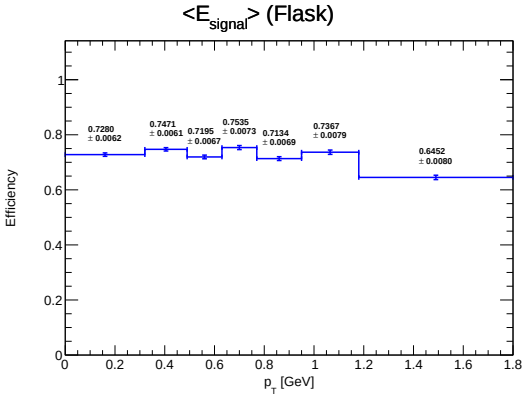


Empty Flask: Final Signal Efficiency (E_final_signal_Flask.pdf)

Existing Results

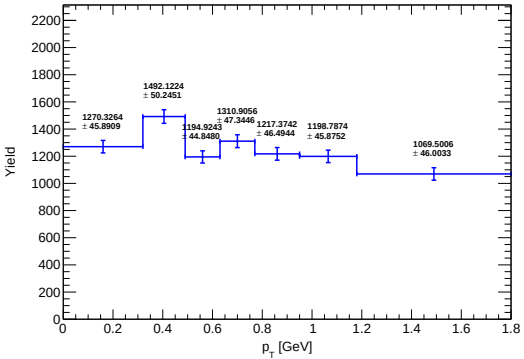


Latest Results

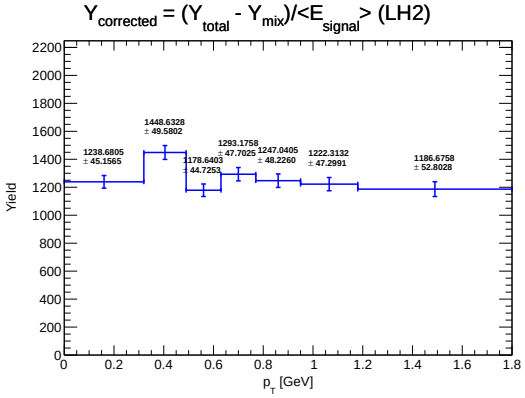


LH_2 Target: Corrected Yield (Y_corrected_LH2.pdf)

Existing Results

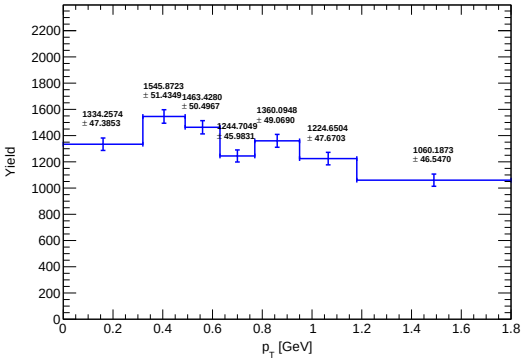


Latest Results

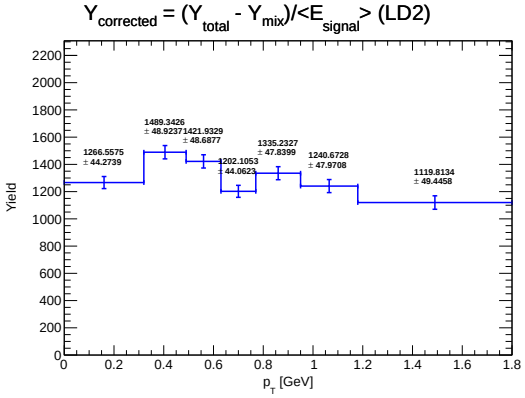


LD_2 Target: Corrected Yield (`Y_corrected_LD2.pdf`)

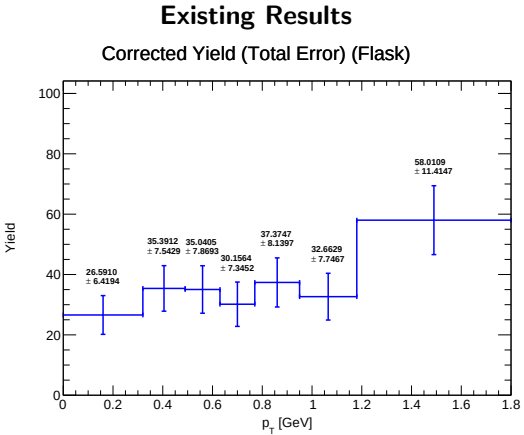
Existing Results

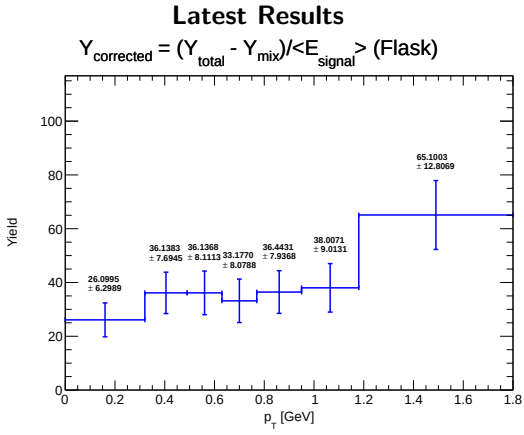


Latest Results



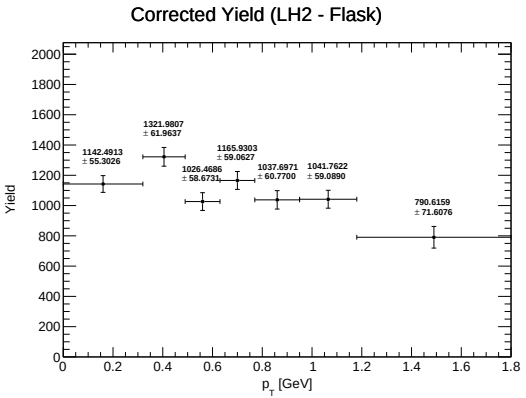
Empty Flask: Corrected Yield (Y_corrected_Flask.pdf)



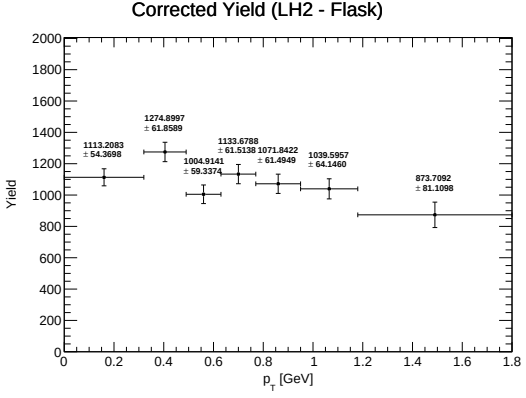


LH_2 Target: Subtracted Corrected Yield (Y_corrected_Subtracted_LH2.pdf)

Existing Results

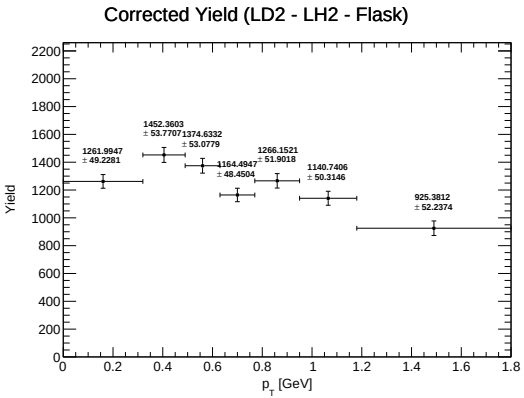


Latest Results

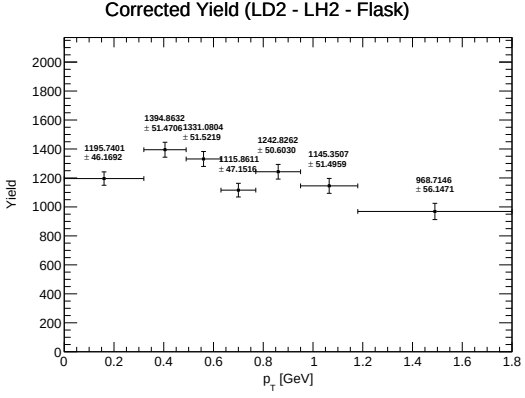


LD_2 Target: Subtracted Corrected Yield (Y_corrected_Subtracted_LD2.pdf)

Existing Results

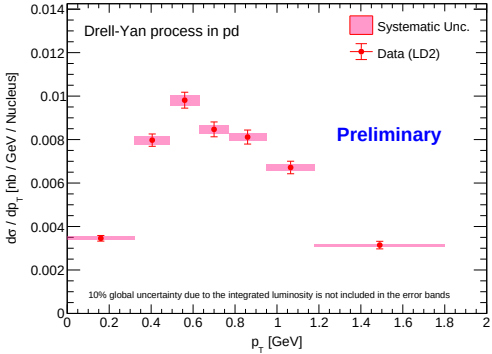


Latest Results

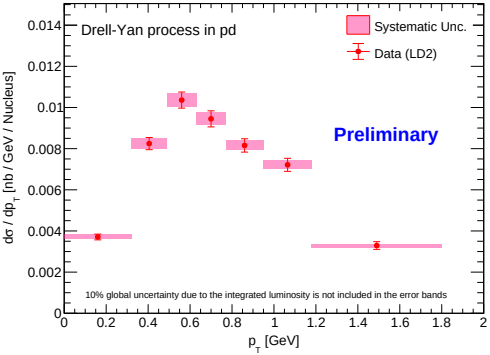


LD_2 Cross Section vs. p_T (CrossSection_LD2_geom_vs_pT.pdf)

Existing Results

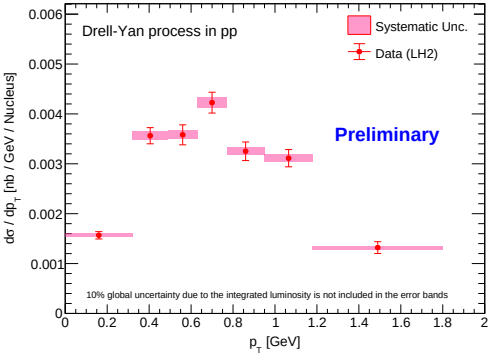


Latest Results

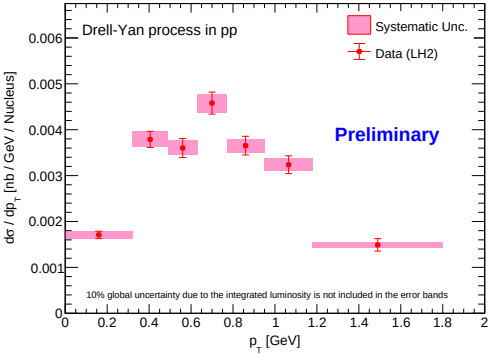


LH_2 Cross Section vs. p_T (CrossSection_LH2_geom_vs_pT.pdf)

Existing Results

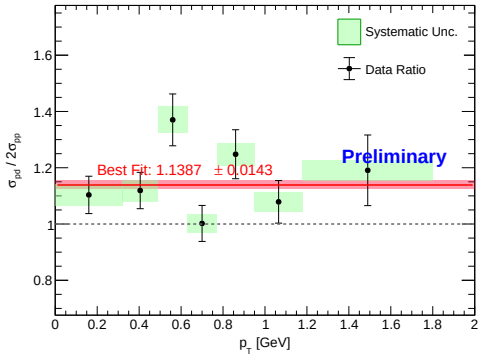


Latest Results

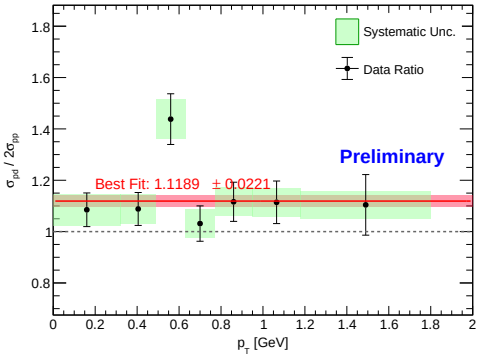


Cross Section Ratio $\sigma_{pd}/2\sigma_{pp}$ vs. p_T (CrossSection_Ratio_pd_2pp_vs_pT_geom.pdf)

Existing Results

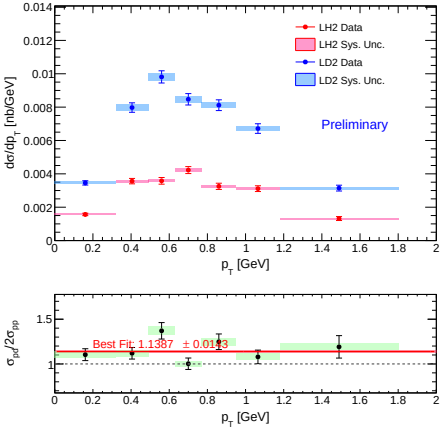


Latest Results

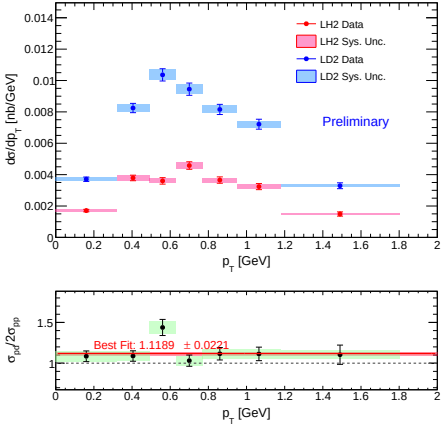


Combined Cross Section & Ratio vs. p_T (Combined_XSec_Ratio_vs_pT_geom.pdf)

Existing Results



Latest Results



LH_2 Target: Systematic Components (Table_Systematics_LH2.tex)

Latest Results

$[p_{T,\min}, p_{T,\max})$	Acc. Corr.	Acc. Err.	Acc. Frac. Err.	ψ' Corr.	ψ' Err.	ψ' Frac. Err.	Yield Corr.	Yield Err.	Yield Frac. Err.
[0.00, 0.32)	0.017346	0.000778	0.044843	0.001829	0.000453	0.247808	1113.2083	14.1447	0.012706
[0.32, 0.49)	0.016862	0.000699	0.041470	0.002900	0.000720	0.248166	1274.8997	16.8204	0.013194
[0.49, 0.63)	0.016977	0.000698	0.041108	0.003874	0.000957	0.247144	1004.9141	13.9272	0.013859
[0.63, 0.77)	0.015061	0.000633	0.042035	0.003034	0.001294	0.426602	1133.6788	15.5546	0.013720
[0.77, 0.95)	0.013884	0.000534	0.038469	0.003310	0.000768	0.232102	1071.8422	15.1637	0.014147
[0.95, 1.18)	0.011893	0.000457	0.038421	0.004258	0.000933	0.219190	1039.5957	14.7739	0.014211
[1.18, 1.80)	0.008053	0.000282	0.035065	0.003692	0.000794	0.215045	873.7092	17.7642	0.020332

Existing Results

$[p_{T,\min}, p_{T,\max})$	Acc. Corr.	Acc. Err.	Acc. Frac. Err.	ψ' Corr.	ψ' Err.	ψ' Frac. Err.	Yield Corr.	Yield Err.	Yield Frac. Err.
[0.00, 0.32)	0.019414	0.000441	0.022716	0.001829	0.000453	0.247808	1142.4913	13.0388	0.011413
[0.32, 0.49)	0.018594	0.000392	0.021087	0.002900	0.000720	0.248166	1321.9807	14.9397	0.011301
[0.49, 0.63)	0.017444	0.000378	0.021680	0.003874	0.000957	0.247144	1026.4686	12.0291	0.011719
[0.63, 0.77)	0.016789	0.000357	0.021237	0.003034	0.001294	0.426602	1165.9303	12.8487	0.011020
[0.77, 0.95)	0.015105	0.000297	0.019683	0.003310	0.000768	0.232102	1037.6971	12.3699	0.011921
[0.95, 1.18)	0.012398	0.000249	0.020057	0.004258	0.000933	0.219190	1041.7622	12.1026	0.011617
[1.18, 1.80)	0.008227	0.000153	0.018561	0.003692	0.000794	0.215045	790.6159	10.2818	0.013005

LD_2 Target: Systematic Components (Table_Systematics_LD2.tex)

Latest Results

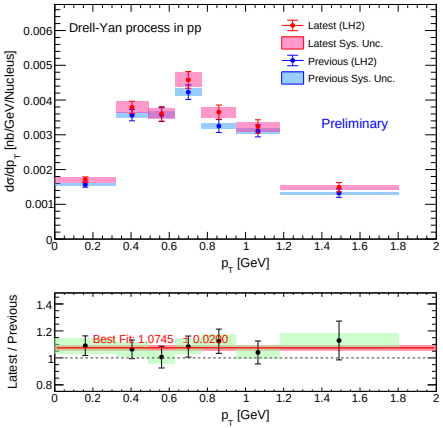
$[p_{T,\min}, p_{T,\max})$	Acc. Corr.	Acc. Err.	Acc. Frac. Err.	ψ' Corr.	ψ' Err.	ψ' Frac. Err.	Yield Corr.	Yield Err.	Yield Frac. Err.
[0.00, 0.32)	0.017935	0.000555	0.030941	0.001829	0.000453	0.247808	1195.7401	9.7708	0.008171
[0.32, 0.49)	0.017707	0.000501	0.028267	0.002900	0.000720	0.248166	1394.8632	11.9731	0.008584
[0.49, 0.63)	0.016331	0.000480	0.029375	0.003874	0.000957	0.247144	1331.0804	11.8553	0.008907
[0.63, 0.77)	0.015013	0.000442	0.029424	0.003034	0.001294	0.426602	1115.8611	9.9159	0.008886
[0.77, 0.95)	0.015065	0.000388	0.025787	0.003310	0.000768	0.232102	1242.8262	11.2123	0.009022
[0.95, 1.18)	0.012283	0.000324	0.026379	0.004258	0.000933	0.219190	1145.3507	10.9334	0.009546
[1.18, 1.80)	0.008446	0.000202	0.023906	0.003692	0.000794	0.215045	968.7146	11.4972	0.011869

Existing Results

$[p_{T,\min}, p_{T,\max})$	Acc. Corr.	Acc. Err.	Acc. Frac. Err.	ψ' Corr.	ψ' Err.	ψ' Frac. Err.	Yield Corr.	Yield Err.	Yield Frac. Err.
[0.00, 0.32)	0.020298	0.000441	0.021727	0.001829	0.000453	0.247808	1261.9947	13.5151	0.010709
[0.32, 0.49)	0.019066	0.000392	0.020565	0.002900	0.000720	0.248166	1452.3603	14.8908	0.010253
[0.49, 0.63)	0.017807	0.000378	0.021237	0.003874	0.000957	0.247144	1374.6332	13.7162	0.009978
[0.63, 0.77)	0.017477	0.000357	0.020401	0.003034	0.001294	0.426602	1164.4947	11.7613	0.010100
[0.77, 0.95)	0.015424	0.000297	0.019276	0.003310	0.000768	0.232102	1266.1521	12.7874	0.010099
[0.95, 1.18)	0.013143	0.000249	0.018919	0.004258	0.000933	0.219190	1140.7406	11.5325	0.010110
[1.18, 1.80)	0.008446	0.000153	0.018081	0.003692	0.000794	0.215045	925.3812	9.6742	0.010454

Cross-Section Comparison vs. p_T: LH_2 and LD_2

LH_2 Plot



LD_2 Plot

